

Step-by-Step Guide:

Draft Your AI Research Proposal Today

Course:

Research Methodology

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Must Have (Comprehensive AI Research Proposal Outline)**1. Title Page** *(required)*

1.1 Full Project Title

1.2 Principal Investigator / Student Name & ID

1.3 Supervisor(s) and Department

1.4 Submission Date

2. Abstract *(required)*

- One-paragraph summary: problem, gap, approach, data, expected results.

3. Table of Contents *(required)***4. Introduction & Motivation** *(required)*

4.1 Context and Real-World Importance

4.2 Core Problem Statement

4.3 Project Goals

4.4 Scope and Delimitations

5. Literature Review *(required)*

5.1 State-of-the-Art Techniques

5.2 Known Limitations / Open Questions

5.3 Comparative Table of Related Work

5.4 Synthesis & Research Gap

6. Problem Statement, Objectives, and Hypotheses *(required)*

6.1 Formal Problem Definition

6.2 Research Questions (**Max 3**)

6.3 Specific, Measurable Objectives

6.4 Testable Hypotheses

- *For each key research question or methodological claim, state a hypothesis that is empirically verifiable (falsifiable).*
- **E.g., H1:** “Model X with domain-adaptive pre-training will increase accuracy on dataset Z by ≥ 10 points over baseline Model Y.”
- **H2:** “Adding Self-Attention will significantly reduce false positives for class W without raising inference time above 120 ms per image.”
- Separate null/alternative hypotheses.

7. Significance & Contributions

7.1 Theoretical Contributions

7.2 Practical Contributions

7.3 Impact Pathways

8. Proposed Methodology

8.1 Overview & Rationale

8.2 Research Design & Workflow

8.2.1 Data Acquisition & Pre-processing

8.2.2 Model Architecture & Novel Components

8.2.3 Training & Implementation

8.3 Datasets & Baselines

8.4 Evaluation Plan

8.4.1 Quantitative Metrics

8.4.2 Qualitative Analysis

8.4.3 Ablation & Sensitivity Studies

8.5 Risk Mitigation & Contingency

8.6 Fairness, Ethics, and Safety

9. Project Management

10.1 Timeline / Gantt Chart

10.2 Roles & Responsibilities

10.3 Budget & Resource Needs

10. References *(required)*

Your Step-by-Step Guide: Draft Your AI Research Proposal Today**10-15 Papers, 3 RQ**

Welcome! The goal of today's session is to guide you from a broad topic to a complete first draft of your research proposal, following a professional academic structure. Follow these steps, manage your time, and you will leave with a powerful head start on your project.

What You'll Need:

- A laptop with internet access and a word processor.
- Access to research databases (e.g., Google Scholar, IEEE Xplore, ACM Digital Library).
- Your ideas for 1-2 broad research topics (e.g., Computer Vision, NLP).

Step 1: Introduction & Topic Narrowing

Goal: To define your research area and draft the initial ideas for your **Section 4: Introduction & Motivation**.

1. Choose & Narrow Your Topic:

- Select one broad topic (e.g., NLP).
- Narrow it down to a specific sub-topic (e.g., "Fact-checking and misinformation detection"). Use the provided class list for inspiration if needed.
- **Action:** In your proposal document, under a temporary heading "My Ideas," write down the context of this problem and why it's important in the real world. This will become **Section 4.1 (Context and Real-World Importance)**.

2. Pair & Share:

- Turn to a partner. Explain your topic and why it matters. Get feedback on its clarity and focus.
- **Action:** Refine your ideas based on the feedback. Write a single, clear sentence that states the core problem you want to address. This is the seed for **Section 4.2 (Core Problem Statement)**.

Step 2: Lightning Literature Review & Gap Finding

Goal: To find a specific, evidence-based research gap and draft **Section 5: Literature Review**.

1. Find Your Papers:

- Go to Google Scholar. Use focused keywords like "misinformation detection survey 2024" or "limitations of BERT for fact-checking".
- Your goal is to find **5-7 recent, relevant papers**. Use the "Three-Pass Reading Strategy" (provided at the end) to quickly assess them.

2. Create a Gap Table:

- For the papers you keep, create a "Gap Table." This will become your **Section 5.3 (Comparative Table of Related Work)**.
- **Action:** Fill out the table below. Pay close attention to the last column.

Author & Year	Method Used	Key Finding / Contribution	Stated Gap / Limitation
(e.g., Li, 2024)	(e.g., RoBERTa)	(e.g., 91% F1-score)	(e.g., "Performs poorly on claims with numerical data.")
Gourav Bathla et al 2022	Hybrid	93%	...

3. Synthesize Your Gap:

- Look for patterns in your table. If multiple papers mention the same limitation, you've found a strong gap.
- Action:** Write a concluding paragraph for your literature review that summarizes your findings and explicitly states the gap. This is **Section 5.4 (Synthesis & Research Gap)**. Example: *"While existing models show high performance, they consistently fail to verify claims involving numerical data. This limitation represents a significant, unaddressed gap..."*

Step 3: Define Your Problem, Objectives & Hypotheses

Goal: To create a sharp, testable project focus by drafting **Section 6**.

1. Formalize Your Problem:

- Action:** Based on your gap, draft the following:
 - Section 6.1 (Formal Problem Definition):** Describe the problem concisely.
 - Section 6.2 (Research Questions):** Write 1-2 specific, answerable questions. (e.g., *"How can numerical reasoning be integrated into a language model to improve fact-checking accuracy?"*).

2. Set Objectives & Hypotheses:

- Action:** For each Research Question, write:
 - Section 6.3 (Specific Objectives):** A bullet point list of what you will do. (e.g., "1. Develop a numerical-aware attention module. 2. Evaluate the model on the ABC dataset.").
 - Section 6.4 (Testable Hypotheses):** Your specific, measurable prediction. Use the H_1 (Alternative Hypothesis) format.
 - Template:** "The proposed [Your Method] will achieve a [Metric] of [Target Value] on [Dataset], representing a significant improvement over the [Baseline Model]."
 - Example (H_1):** *"The proposed numerical-aware model (H_1) will increase the F1-score on the NumFact dataset by at least 10 points compared to the baseline RoBERTa model."*
 - Note the Null Hypothesis (H_0):** *"There will be no significant difference in F1-score..."* Your goal is to reject H_0 .

3. Peer Review: Quickly check with a partner: Are your RQs and hypotheses clear, specific, and measurable?

Step 4: Draft Methodology & Significance

Goal: To outline *how* you'll do the work (**Section 8**) and *why* it matters (**Section 7**).

1. Draft Your Methodology:

- **Action:** Write brief, bullet-point notes for each part of **Section 8 (Proposed Methodology)**.
 - **8.1 Overview & Rationale:** (Why is your proposed method a good idea for this specific gap?)
 - **8.2.1 Data Acquisition:** (What dataset will you use? Is it public?)
 - **8.2.2 Model Architecture:** (What is your core idea? A new layer? A new loss function?)
 - **8.3 Datasets & Baselines:** (What models will you compare against?)
 - **8.4.1 Quantitative Metrics:** (How will you measure success? Accuracy, F1-score, etc.)
 - **8.6 Fairness, Ethics, or Safety:** (Any potential negative impacts to consider?)

2. State Your Contributions:

- **Action:** Draft **Section 7 (Significance & Contributions)**. Answer briefly:
 - **7.1 Theoretical:** (Will you propose a new technique or theory?)
 - **7.2 Practical:** (Will you create a new dataset, tool, or solve a real-world problem?)

Step 5: Plan Your Project & Assemble the Draft (30 minutes)

Goal: To create a realistic timeline and compile all your work into a single document.

1. Create a Simple Timeline:

- **Action:** Draft **Section 9.1 (Timeline)**. Break your project into 4-5 major phases and estimate the weeks needed for each.
 - *Phase 1: In-depth Literature Review & Data Prep (Weeks 1-3)*
 - *Phase 2: Model Implementation & Prototyping (Weeks 4-7)*
 - *Phase 3: Main Experiments & Analysis (Weeks 8-10)*
 - *Phase 4: Final Write-up & Submission (Weeks 11-12)*

2. Assemble Your Full Proposal:

- Create a new document using the full proposal outline as your template.
- Copy and paste your drafted notes from each step into the correct sections.
- Fill in the **Title Page (Section 1)**.
- Create a **References (Section 10)** list with the papers from your Gap Table.
- **Read through the entire document.** Does it flow logically? Refine the language to connect the sections.

3. Final Action:

- Write the **Abstract (Section 2)** *last*. It should be a one-paragraph summary of your completed proposal.
- Generate the **Table of Contents (Section 3)** using your word processor's built-in tool.

Congratulations! You now have a comprehensive first draft of your research proposal.